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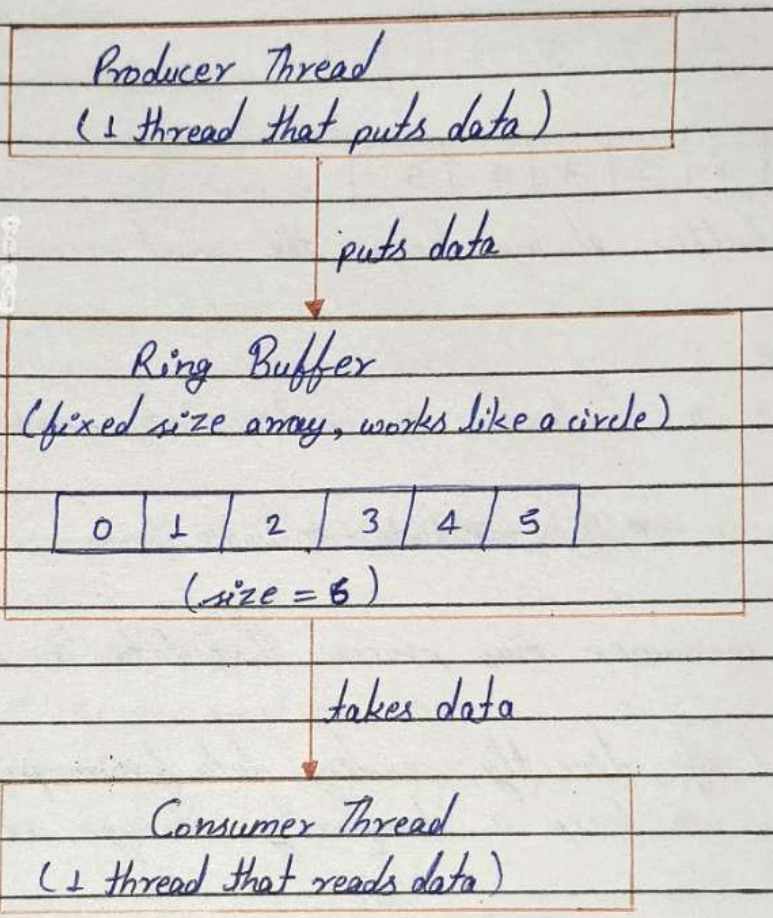
SPSC Ring-Buffer :-

SPSC → Single Producer, Single Consumer

→ It means there are only two threads:

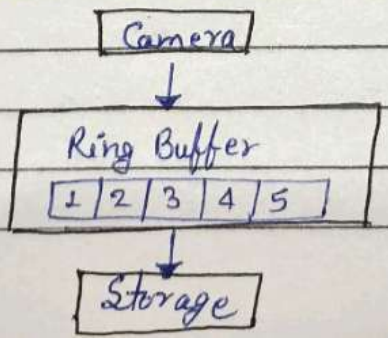
Producer: puts data into the buffer.

Consumer: takes data from the buffer.



Ring Buffer : A ring buffer is a fixed size array that works like a circle.

Example : Imagine a CCTV camera has a small storage buffer that can hold the last 5 video chunks:



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Suppose the camera keeps receiving video:

1 → 2 → 3 → 4 → 5

The buffer becomes full.

When new video 6 arrives, instead of creating more space, it overwrites the oldest chunks.

Before:

1	2	3	4	5

After:

6	2	3	4	5	.....

So the buffer always keeps the most recent data.

→ Why need a Ring Buffer

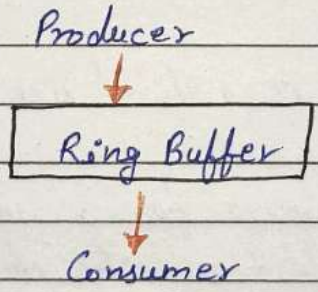
Suppose a producer is continuously creating data:

Producer → Data → Data → Data -----

The consumer may process that data a little later.

So,

Instead of directly sending data from producer to consumer, we keep a temporary storage area:



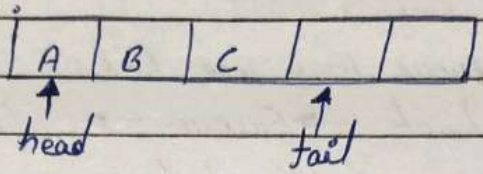
This is useful when the producer and consumer work at different speeds.

→ Here, two important variables are used:
head and tail

head: head tells us where the consumer will read the next item.

tail: tail tells us where the producer will write the next item.

Example: Suppose:



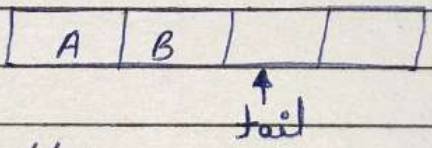
So, consumer reads A while producer can write the next item at tail position.

→ How producer works

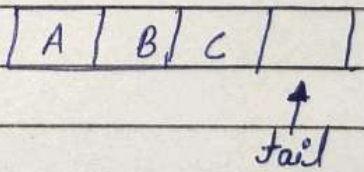
The producer follows roughly these steps:

* Check if there is space
* Put data into the buffer
* Move tail to the next position

Example: Before:



Producer add C:



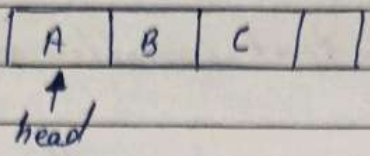
→ How consumer works

The consumer does the opposite such as

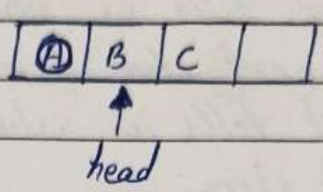
* Check if data is available
* Read the data
* Move head to the next position

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Example: Before :



Consumer read A :



→ Why SPSC Ring Buffer fast

* A traditional queue may use locks :
  Producer → Lock → Queue → Unlock
* The consumer also needs the lock.
* Locks can cause threads to wait.
* An SPSC ring buffer can be designed without traditional locks :
  Producer → Ring Buffer → Consumer
  This is called lock free design when the implementation satisfies the appropriate lock free progress guarantees.